SCOOT group quizzes

Ten multiple-choice questions for each of the six investigation groups in the SCOOT Foundation lesson. Use them after the group report-back, as homework, or as a quick check before the next lesson. Each quiz takes about 10 minutes. Answers with short explanations are at the end.

Group	Topic	Builds on
Group 1	Small nudges (the elastic plan)	F1 · F6 · F7 · F8
Group 2	Detectors and LPUs	F2 · F3
Group 3	Platoons spread	F2 · F3 · F7
Group 4	Queue and degree of saturation	F1 · F3 · F4 · F5 · F6
Group 5	Validation	F2 · F4 · F6
Group 6	SCOOT vs others	F1 · F5 · F8

Interactive versions with instant feedback: smrahmanbd.github.io/scoot-learning-lab/quiz.html

GROUP 1 QUIZ · 10 QUESTIONS

Small nudges (the elastic plan)

Names: _____ Score: ____ / 10

1. **What are the three timing "dials" in SCOOT's name?**

   A ☐ Speed, flow and density
   B ☐ Phase, stage and link
   C ☐ Red, amber and green
   D ☐ Split, cycle and offset

2. **What is the split?**

   A ☐ How the green time is shared between the stages at a junction
   B ☐ The time to run every stage once
   C ☐ The amber plus all-red time
   D ☐ The time between greens at neighbouring junctions

3. **What is the offset?**

   A ☐ The difference between two splits
   B ☐ The time a car takes to cross the junction
   C ☐ The length of the amber
   D ☐ The time from green at one junction to green at the next one

4. **How often does SCOOT's split optimiser make a decision?**

   A ☐ Once an hour
   B ☐ Every 5 minutes
   C ☐ Once a day
   D ☐ Before every stage change

5. **How often is the cycle time of a region normally reviewed?**

   A ☐ Every second
   B ☐ At every stage change
   C ☐ Every 5 minutes (2.5 when traffic is rising)
   D ☐ Once a week

6. **A junction has two stages and a 60 s cycle. How many split decisions does SCOOT make in one hour?**

   A ☐ 120
   B ☐ 60
   C ☐ 12
   D ☐ 192

7. **Add 120 split decisions, 60 offset decisions (one per cycle) and 12 cycle reviews (every 5 minutes). How many decisions per hour?**

   A ☐ 192
   B ☐ 240
   C ☐ 180
   D ☐ 132

8. **Why does SCOOT change the timings by only a few seconds at a time?**

   A ☐ So drivers do not notice the change
   B ☐ The controller cannot make bigger changes
   C ☐ One wrong guess cannot do much harm, and many small nudges follow the traffic smoothly
   D ☐ To save electricity

9. **Which everyday object is the best picture of the elastic plan?**

   A ☐ A thermostat that nudges the heating every few minutes
   B ☐ A light switch
   C ☐ A printed bus timetable
   D ☐ A road map

10. **How is SCOOT's elastic plan different from "plan selection"?**

    A ☐ SCOOT lets every junction pick its own cycle freely
    B ☐ SCOOT stretches the amber when traffic is heavy
    C ☐ SCOOT edits the running plan a little at a time instead of swapping whole stored plans
    D ☐ SCOOT swaps between stored plans at fixed times of day

GROUP 2 QUIZ · 10 QUESTIONS

Detectors and LPUs

Names: _____ Score: ____ / 10

1. **Where is the SCOOT detector loop normally placed?**

   A ☐ Near the far end of the link, just after the upstream junction

   B ☐ In the middle of the junction

   C ☐ At the stop line

   D ☐ On the exit road

2. **What does SCOOT gain from putting the loop there?**

   A ☐ About 10–20 s of warning before the cars reach the stop line

   B ☐ A count of pedestrians

   C ☐ A cheaper installation

   D ☐ An exact speed for every car

3. **How often is a SCOOT loop read?**

   A ☐ Once a second

   B ☐ Once a cycle

   C ☐ 4 times a second

   D ☐ Every 5 minutes

4. **What does LPU stand for?**

   A ☐ Lane priority unit

   B ☐ Link profile unit

   C ☐ Loop pulse unit

   D ☐ Lost period unit

5. **Which rule does this lab use to turn a vehicle into LPUs?**

   A ☐ LPU = 1 × vehicle + 2 × seconds the loop is covered

   B ☐ LPU = vehicles ÷ seconds

   C ☐ LPU = seconds covered only

   D ☐ LPU = 2 × vehicle + 1 × seconds covered

6. **A 4.5 m car crosses a 2 m loop at 13.5 m/s. How long is the loop covered?**

   A ☐ 1.04 s

   B ☐ 0.33 s

   C ☐ 0.48 s

   D ☐ 2.25 s

7. **Using the lab's rule, how many LPUs is that car?**

   A ☐ 1.00

   B ☐ 2.96

   C ☐ 1.96

   D ☐ 1.48

8. **A 12 m bus crosses the same loop at the same speed. How many LPUs?**

   A ☐ 1.04

   B ☐ 4.00

   C ☐ 2.07

   D ☐ 3.07

9. **In a jam, a car crawls slowly over the loop. What happens to its LPU value?**

   A ☐ It stays exactly the same

   B ☐ It rises, because the loop is covered for longer

   C ☐ It falls

   D ☐ It becomes zero

10. **Why does SCOOT use LPUs instead of plain vehicle counts?**

    A ☐ The law requires them

    B ☐ LPUs reflect how much green the traffic needs: long or slow vehicles count more

    C ☐ LPUs ignore buses

    D ☐ LPUs are easier to count

GROUP 3 QUIZ · 10 QUESTIONS

Platoons spread

Names: _____ Score: ____ / 10

1. **What is a platoon?**
   - **A** ☐ The queue at the stop line
   - **B** ☐ A single lane of a road
   - **C** ☐ A bunch of cars released together by the upstream green
   - **D** ☐ A convoy of buses only

2. **What is a cyclic flow profile?**
   - **A** ☐ The speed of every car
   - **B** ☐ A chart of traffic over a whole day
   - **C** ☐ The order of the stages
   - **D** ☐ The LPUs seen in each second of the cycle, averaged over several cycles

3. **Why does a platoon spread out on its way to the stop line?**
   - **A** ☐ The detector misses some cars
   - **B** ☐ Drivers travel at different speeds
   - **C** ☐ The road gets narrower
   - **D** ☐ The signals change colour

4. **What is the cruise time T?**
   - **A** ☐ The cycle time
   - **B** ☐ The time to clear the queue
   - **C** ☐ The length of the green
   - **D** ☐ The time to drive from the detector to the stop line

5. **With T = 10 s and β = 0.8, when do the first predicted arrivals reach the stop line?**
   - **A** ☐ After 10 s
   - **B** ☐ After 2 s
   - **C** ☐ After 8 s
   - **D** ☐ After 12.5 s

6. **With α = 0.35, β = 0.8 and T = 10 s, what is the smoothing factor F = 1 ÷ (1 + αβT)?**
   - **A** ☐ 0.263
   - **B** ☐ 0.50
   - **C** ☐ 0.10
   - **D** ☐ 0.80

7. **What does F = 0.263 mean?**
   - **A** ☐ The platoon arrives 26% faster
   - **B** ☐ 26% of each second's traffic arrives first; 74% spills into later seconds
   - **C** ☐ 26% of the cycle is green
   - **D** ☐ 26% of the cars are lost

8. **On a longer road with T = 20 s, F becomes...**
   - **A** ☐ Exactly the same
   - **B** ☐ Zero
   - **C** ☐ Larger, so the platoon spreads less
   - **D** ☐ Smaller (about 0.15), so the platoon spreads more

9. **A steady 1 LPU/s arrives at the loop. Second 1 at the stop line is 0.263. What is second 2?**
   - **A** ☐ 0.263
   - **B** ☐ 0.526
   - **C** ☐ 0.737
   - **D** ☐ 0.457

10. **Compared with the platoon at the detector, the platoon at the stop line has...**
    - **A** ☐ More vehicles
    - **B** ☐ The same vehicles, a lower peak and a longer duration
    - **C** ☐ Fewer vehicles
    - **D** ☐ Exactly the same shape

GROUP 4 QUIZ · 10 QUESTIONS

Queue and degree of saturation

Names: _____ Score: ____ / 10

1. **In the bathtub picture of a queue, what is the tap?**
   - **A** ☐ The arriving traffic
   - **B** ☐ The queue
   - **C** ☐ The saturation occupancy
   - **D** ☐ The green light

2. **What is SATO?**
   - **A** ☐ Saturation occupancy: how fast a standing queue drains in green
   - **B** ☐ The average speed on the link
   - **C** ☐ The amber time
   - **D** ☐ The offset between junctions

3. **Arrivals are 0.2 LPU/s and red lasts 35 s. How big is the queue at the end of red?**
   - **A** ☐ 7 LPU
   - **B** ☐ 3.5 LPU
   - **C** ☐ 12 LPU
   - **D** ☐ 5 LPU

4. **In green, SATO is 0.6 LPU/s and cars still arrive at 0.2 LPU/s. How fast does the queue shrink?**
   - **A** ☐ 0.2 LPU/s
   - **B** ☐ 0.8 LPU/s
   - **C** ☐ 0.4 LPU/s
   - **D** ☐ 0.6 LPU/s

5. **How long does a 7 LPU queue take to clear at 0.4 LPU/s?**
   - **A** ☐ 17.5 s
   - **B** ☐ 7 s
   - **C** ☐ 12 s
   - **D** ☐ 28 s

6. **What is the degree of saturation (DoS)?**
   - **A** ☐ Delay ÷ stops
   - **B** ☐ Queue × SATO
   - **C** ☐ Green time ÷ cycle time
   - **D** ☐ Demand ÷ what the green can clear

7. **Demand is 0.2 × 60 = 12 LPU per cycle and the green can clear 25 × 0.6 = 15 LPU. What is the DoS?**
   - **A** ☐ 0.80
   - **B** ☐ 1.25
   - **C** ☐ 1.50
   - **D** ☐ 0.60

8. **Why does SCOOT aim for about 90% rather than 100%?**
   - **A** ☐ To save fuel
   - **B** ☐ Because detectors fail above 90%
   - **C** ☐ To leave a margin for random surges in traffic
   - **D** ☐ Because 100% is illegal

9. **A link has a DoS of 1.10. What happens?**
   - **A** ☐ The queue grows every cycle
   - **B** ☐ There is plenty of spare green
   - **C** ☐ The green is exactly used up
   - **D** ☐ There is no traffic

10. **In the queue model, total delay equals...**
    - **A** ☐ The number of stops
    - **B** ☐ The area under the queue-over-time line
    - **C** ☐ SATO × green time
    - **D** ☐ The length of the red

GROUP 5 QUIZ · 10 QUESTIONS

Validation

Names: _____ Score: ____ / 10

1. **What is validation in SCOOT?**

   A ☐ Tuning each link's model so it matches what happens on the street

   B ☐ Counting accidents

   C ☐ Installing new loops

   D ☐ Checking the signal bulbs

2. **The model and the engineer's stopwatch disagree about when the queue clears. Which is right?**

   A ☐ Neither

   B ☐ The street (the stopwatch)

   C ☐ The average of the two

   D ☐ The model

3. **Which formula gives SATO from a stopwatch measurement?**

   A ☐ SATO = clear time ÷ queue

   B ☐ SATO = (clear time + start lag) ÷ queue

   C ☐ SATO = queue ÷ (clear time − start lag)

   D ☐ SATO = queue × start lag

4. **A 19.5 LPU queue clears 36 s after green starts; the start lag is 3 s. What is SATO?**

   A ☐ 0.54 LPU/s

   B ☐ 0.67 LPU/s

   C ☐ 0.59 LPU/s

   D ☐ 0.81 LPU/s

5. **With SATO set to 0.67 LPU/s, when does the model say that 19.5 LPU queue clears (start lag 3 s)?**

   A ☐ After about 29 s

   B ☐ After about 32 s

   C ☐ After 36 s

   D ☐ After 39 s

6. **What goes wrong if SATO is set too high?**

   A ☐ SCOOT stops working

   B ☐ The model thinks queues clear sooner, gives too little green, and real queues grow

   C ☐ The cycle doubles

   D ☐ SCOOT gives far too much green

7. **What is the start lag?**

   A ☐ The travel time between junctions

   B ☐ The amber time

   C ☐ The time drivers take to get going after the green starts

   D ☐ The length of the cycle

8. **Which of these is NOT a setting checked during validation?**

   A ☐ The colour of the signal heads

   B ☐ Journey (cruise) time

   C ☐ Start and end lags

   D ☐ Saturation occupancy (SATO)

9. **Which street features can lower SATO?**

   A ☐ Narrow lanes, uphill gradients, parking and turning traffic

   B ☐ Wide straight lanes

   C ☐ Extra detectors

   D ☐ Freshly painted markings

10. **Which analogy fits validation best?**

    A ☐ Reading a map

    B ☐ Buying a new car

    C ☐ Painting a road

    D ☐ Tuning a guitar by ear until it matches a tuning fork

GROUP 6 QUIZ · 10 QUESTIONS

SCOOT vs others

Names: _____ Score: ____ / 10

1. **How does a fixed-time signal decide its green?**
   - **A** ☐ It follows a stored plan
   - **B** ☐ It uses stop-line detectors
   - **C** ☐ It waits for pedestrians
   - **D** ☐ It runs a model fed by upstream detectors

2. **How does an actuated signal decide its green?**
   - **A** ☐ It copies its neighbour
   - **B** ☐ It runs a model of the whole region
   - **C** ☐ It extends the green while cars keep arriving at stop-line loops, and ends it on a gap
   - **D** ☐ It never changes

3. **Which description fits SCOOT?**
   - **A** ☐ Upstream loops feed a model; small frequent changes; coordinated junctions
   - **B** ☐ A stored plan only
   - **C** ☐ Random timings
   - **D** ☐ One loop at the stop line, no model

4. **In Lab F3 (one junction in SUMO), which controller had the lowest mean delay?**
   - **A** ☐ All the same
   - **B** ☐ Fixed time
   - **C** ☐ SCOOT
   - **D** ☐ Actuated

5. **SCOOT 12.79 s against fixed time 18.96 s. About how much less delay is that?**
   - **A** ☐ 33%
   - **B** ☐ 25%
   - **C** ☐ 50%
   - **D** ☐ 12%

6. **SCOOT 12.79 s against actuated 14.48 s. About how much less delay?**
   - **A** ☐ 5%
   - **B** ☐ 33%
   - **C** ☐ 20%
   - **D** ☐ 12%

7. **In the four-junction corridor run, which controller had slightly less mean delay?**
   - **A** ☐ All equal
   - **B** ☐ Fixed time
   - **C** ☐ SCOOT
   - **D** ☐ Actuated (29.1 s vs SCOOT 30.4 s)

8. **In that corridor run, what was SCOOT's clearest advantage?**
   - **A** ☐ It used the shortest cycle
   - **B** ☐ Fewer stops and much less bus delay
   - **C** ☐ It was cheapest
   - **D** ☐ It needed no detectors

9. **Where might a simple fixed or actuated controller be good enough?**
   - **A** ☐ On a corridor with closely spaced junctions
   - **B** ☐ In a busy city-centre grid
   - **C** ☐ Where many buses need priority
   - **D** ☐ At an isolated, lightly used junction

10. **What does SCOOT need in order to work well?**
    - **A** ☐ A fixed cycle
    - **B** ☐ Working detectors and a validated model
    - **C** ☐ GPS in every car
    - **D** ☐ No pedestrians

Answer key

Group 1 · Small nudges (the elastic plan)		
1	**D**	SCOOT = Split Cycle Offset Optimisation Technique: it adjusts those three timing variables.
2	**A**	The split shares the green; the cycle is the full round of stages; the offset links neighbouring junctions.
3	**D**	Offsets coordinate neighbours so a platoon released by one junction meets a green at the next (a green wave).
4	**D**	A few seconds before each stage change it tries changing it a little earlier, keeping it, or a little later.
5	**C**	The cycle optimiser works on a slower clock than the split and offset optimisers.
6	**A**	3600 ÷ 60 = 60 cycles; two stage changes per cycle; 60 × 2 = 120.
7	**A**	120 + 60 + 12 = 192 small timing decisions an hour at one junction.
8	**C**	Small, frequent changes are robust: errors stay small and the plan tracks demand like a thermostat.
9	**A**	A thermostat makes small corrections often. A printed timetable is the picture of a fixed-time plan.
10	**C**	Plan selection jumps between pre-made plans; SCOOT keeps one plan and stretches it gently, like elastic.

Group 2 · Detectors and LPUs		
1	**A**	Upstream placement means cars are counted long before they reach the stop line.
2	**A**	The warning lets the model predict arrivals and get the green ready in time.
3	**C**	Each quarter second the loop reports whether it is covered.
4	**B**	LPUs are SCOOT's own unit for an amount of traffic, mixing a count with occupancy.
5	**A**	One unit for being a vehicle, plus two for every second it blocks the loop. (The real TRL coding differs; this is the lab's teaching rule.)
6	**C**	Covered time = (vehicle length + loop length) ÷ speed = 6.5 ÷ 13.5 = 0.48 s.
7	**C**	1 + 2 × 0.48 = 1.96 LPU.
8	**D**	Covered (12 + 2) ÷ 13.5 = 1.04 s, so 1 + 2 × 1.04 = 3.07 LPU.
9	**B**	Slow traffic covers the loop longer, so LPUs rise: an early sign of congestion.
10	**B**	A bus needs more green than a car; counting both as "1" would hide that.

Group 3 · Platoons spread		
1	**C**	When the upstream light turns green, cars leave together as a platoon.
2	**D**	One bar per second of the cycle; the platoon shows up as a block of tall bars.
3	**B**	Fast drivers pull ahead and slow drivers fall behind, so the bunch gets longer and flatter.
4	**D**	It is measured on the street during validation.
5	**C**	Lag = β × T = 0.8 × 10 = 8 s: the fastest drivers arrive before the average car.
6	**A**	1 ÷ (1 + 0.35 × 0.8 × 10) = 1 ÷ 3.8 = 0.263.
7	**B**	Each second: arrivals = 0.263 × new traffic + 0.737 × last second's arrivals. Nobody is lost.
8	**D**	1 ÷ (1 + 0.35 × 0.8 × 20) = 1 ÷ 6.6 ≈ 0.15. The longer the trip, the more the bunch smears out.
9	**D**	0.263 × 1 + 0.737 × 0.263 = 0.263 + 0.194 = 0.457.

10	**B**	The total is unchanged (10 LPU in the example); it is only rearranged in time, over about 25 s instead of 10 s.

Group 4 · Queue and degree of saturation		
1	**A**	Arrivals pour in all the time; red is the plug; the plughole is the green discharge (SATO).
2	**A**	It is measured in LPU per second of green, and checked on site during validation.
3	**A**	0.2 × 35 = 7 LPU.
4	**C**	Net drain = 0.6 − 0.2 = 0.4 LPU/s.
5	**A**	7 ÷ 0.4 = 17.5 s after the green starts.
6	**D**	DoS = (arrivals per cycle + leftover queue) ÷ (effective green × SATO).
7	**A**	12 ÷ 15 = 0.80 (80%): below the 90% target, so there is spare green.
8	**C**	Traffic varies from cycle to cycle; a 10% margin stops a single busy cycle from building a queue.
9	**A**	Above 1.0 more arrives than the green can clear, so the leftover queue grows each cycle.
10	**B**	Every vehicle-second spent waiting adds to the area; in the example it is 183.75 LPU·s per cycle.

Group 5 · Validation		
1	**A**	The model is only as good as its settings, and every street is different.
2	**B**	The street is always right: the model is changed to match it.
3	**C**	Take away the start lag, then divide the queue by the time it took to drain.
4	**C**	19.5 ÷ (36 − 3) = 19.5 ÷ 33 = 0.59 LPU/s.
5	**B**	3 + 19.5 ÷ 0.67 = 3 + 29.1 = 32.1 s: 4 s too early compared with the street.
6	**B**	An optimistic plughole means the model under-estimates how much green the link needs.
7	**C**	The first cars react and accelerate before the queue flows at full rate.
8	**A**	Engineers also check the maximum queue clear time on long links.
9	**A**	Anything that slows the queue leaving the stop line lowers the drain rate.
10	**D**	A well-built guitar still plays wrong notes until it is tuned; a well-built model needs tuning to its street.

Group 6 · SCOOT vs others		
1	**A**	Fixed-time plans are often prepared off-line, for example with Webster's formula or TRANSYT.
2	**C**	Actuation reacts to the traffic in front of it, but only locally.
3	**A**	SCOOT predicts from upstream and adapts split, cycle and offset across a region.
4	**C**	Fixed 18.96 s, actuated 14.48 s, SCOOT 12.79 s per vehicle.
5	**A**	(18.96 − 12.79) ÷ 18.96 = 6.17 ÷ 18.96 ≈ 0.33.
6	**D**	(14.48 − 12.79) ÷ 14.48 = 1.69 ÷ 14.48 ≈ 0.12.
7	**D**	SCOOT is usually better, not always: here actuation edged it on mean delay.
8	**B**	Stops 1.10 vs 1.15 per vehicle, and bus delay 22.9 s vs 38.0 s.
9	**D**	Adaptive control pays off most where traffic changes and junctions interact.
10	**B**	With faulty loops or untuned settings the model's predictions — and the timings — go wrong.